PHYS 435 · Exam II · FRONT · A how-to · B basics · C multipoles · D P,D · E capacitors · F BVPs · G images · H math

$k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \cdot \epsilon_0 = 8.85 \times 10^{-12} \cdot \epsilon = \epsilon_0\epsilon_r = \epsilon_0(1+\chi_e)$

FIND IT FAST: problem has... → box

a field (E, P, D) is given	**A5 B1**
charges given: find E or V	**B2 B3**
dielectric + free charge (D, E, P, σ_b)	**A2 D1 D3**
frozen-in P	**D2**
capacitor / coax with dielectric	**E1 E2**
energy, force on dielectric	**E3**
far away; dipole torque/force	**C1 C3 C2**
sphere/cylinder in uniform E₀	**A3 F3**
V given on boundaries	**A3 F2 F6 F7**
charge near plane/sphere/dielectric	**A4 G1 F4**
holes, cavities, overlaps	**B4 D2**
interfaces	**F1**
div/curl, Legendre, integrals	**H1 H2 H4 H5**
HW problem by number	**I1**

★ = most used. Codes: letter = topic group, number = box.

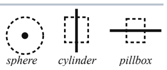
A · HOW TO SOLVE · electric

A1 ★ Which method? (electric)

charge given, high symmetry	Gauss ($\mathbf{D}$ if dielectric)
no symmetry, nearby	Coulomb $\int \frac{k\,dq}{r^2}$, $dq = \lambda dl, \sigma da, \rho d\tau$
no symmetry, far away	multipole: first nonzero of Q, p, quad
V given on boundaries, ρ=0	Laplace + separation
charge + flat/round conductor	images
charge + flat dielectric	partial image $\frac{\epsilon_r-1}{\epsilon_r+1}$
object in uniform E_0	ℓ=1 only: $Ar\cos\theta$, $\frac{B\cos\theta}{r^2}$
a field is given	div/curl → sources
energy	½QV · ½∫D·E · ε₀/2∫E²
holes, overlaps	superpose full shapes

A2 ★ Recipe: Gauss & D → E → P

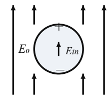
1. **Symmetry sentence:** "by



spherical/cyl./planar symmetry $\mathbf{D} = D(r)\hat{r}$".
2. **Surface** through the point: sphere $4\pi r^2$ · cylinder $2\pi s L$ · pillbox A (2A for a free sheet).
3. D·area $= Q_{f,enc}$; nonuniform: $\int \rho\, 4\pi r'^2 dr'$, $\int \rho\, 2\pi s' L\, ds'$.
4. $E = D/\epsilon$ per region → $P = D - \epsilon_0 E$ → $\sigma_b = \mathbf{P}\cdot\hat{n}$, $\rho_b = -\nabla\cdot\mathbf{P}$.
5. V from ∞ inward, one region at a time.
6. **Check:** $\sum q_b = 0$; E cont. where σ=0; εr→1.

No dielectric: same with $\epsilon_0 E$ and total Q.

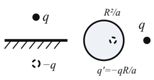
A3 ★ Recipe: Laplace boundary-value problem



1. Regions + coordinates (sphere r,θ · cyl s,φ · box x,y).
2. General solution in each region (F-group).
3. **Kill terms:** center finite ⇒ no $r^{-(l+1)}$, s^{-n}, $\ln s$; ∞ ⇒ no r^l (except $-E_0 r\cos\theta$).
4. **Number BCs:** (i) V cont. (ii) $\epsilon\partial_n V$ jump $= -\sigma_f$ (iii) V on conductor (iv) far field.
5. Expand given $V_0(\theta)$/σ(θ) in P_l; match term by term.
6. $\mathbf{E} = -\nabla V$; σ; check limits (εr→1, ∞).

Uniform field ⇒ only l=1 (n=1). Polynomial in cosθ ⇒ finite l.

A4 ★ Recipe: images



1. Put image charges **outside** the region you solve in.
2. Check V on every boundary (0 or V_0) and at ∞ ⇒ uniqueness ⇒ done.
3. V = real + images; $\sigma = -\epsilon_0\partial_n V$; induced total = Σ images (grounded).
4. Force on q: Coulomb from images only. Energy: ½(q·V of images).

Twists: plane at V_0: +V₀. Sphere at V_0: + $4\pi\epsilon_0 R V_0$ at center. Neutral sphere: + $\frac{qR}{a}$ at center. Dielectric: × $\frac{\epsilon_r-1}{\epsilon_r+1}$, below use q''.

A5 ★ Recipe: given a field → everything

1. Read components + variables; use the one-liner (H-group).
2. $\rho = \epsilon_0\nabla\cdot\mathbf{E}$ (or $\rho_f = \nabla\cdot\mathbf{D}$, $\rho_b = -\nabla\cdot\mathbf{P}$). Electrostatic needs $\nabla\times\mathbf{E} = 0$.
3. Jumps at R: $\sigma = \epsilon_0(E^\perp_{out} - E^\perp_{in})$.
4. Delta: $Q_{enc} = 4\pi\epsilon_0 r^2 E_r$ as r→0 ≠ 0 ⇒ point charge.
5. V: integrate in from ∞; verify $-\nabla V$.
6. Energy: shells; outside a sphere $\frac{Q^2}{8\pi\epsilon_0 R}$; with dielectric ½∫D²/ε.

B · Electrostatics basics (Exam I)

B1 EXAM I · electrostatics core

$\mathbf{E} = k\int \frac{da\,\hat{r}}{r^2} \cdot V = k\int\frac{da}{r} \cdot \mathbf{E} = -\nabla V \cdot V(b) - V(a) = -\int_a^b \mathbf{E}\cdot d\mathbf{l} \cdot \nabla\cdot\mathbf{E} = \frac{\rho}{\epsilon_0} \cdot \nabla\times\mathbf{E} = 0 \cdot \nabla^2 V = -\frac{\rho}{\epsilon_0} \cdot W = q\Delta V$

BC: $E^\perp_{out} - E^\perp_{in} = \frac{\sigma}{\epsilon_0}$, $E^\parallel$ cont., V cont.; $\partial_n V_{out} - \partial_n V_{in} = -\frac{\sigma}{\epsilon_0}$.

Conductors: E=0 inside, V const, charge on surface; just outside $E = \frac{\sigma}{\epsilon_0}\hat{n}$; $\sigma = -\epsilon_0\partial_n V$; pressure $\frac{\sigma^2}{2\epsilon_0}$. Empty cavity: E=0. q in cavity: $-q$ on wall, $+q$(+Q) outside, outside field indep. of q's position. Grounded: V=0; isolated: Q fixed, V unknown const.

Given E → ρ: $\rho = \epsilon_0\nabla\cdot\mathbf{E}$; $Q_{enc}(r) = 4\pi\epsilon_0 r^2 E_r$; $Q_{enc} \to q \neq 0$ as $r \to 0 \Rightarrow q\delta^3$ at origin; E jump at R ⇒ $\sigma = \epsilon_0(E_{out} - E_{in})$. $\int f\delta(x-a) = f(a)$, $\delta(kx) = \frac{\delta(x)}{|k|}$, $\delta^3(\mathbf{r}) = \frac{\delta(r)}{4\pi r^2}$. Ex: $E = \frac{A}{r^2}$ in, $\frac{AR^2}{r^4}$ out ⇒ $4\pi\epsilon_0 A\,\delta^3$, $\rho_{out} = -\frac{2\epsilon_0 AR^2}{r^5}$, $V_{in} = \frac{A}{r} - \frac{2A}{3R}$, $V_{out} = \frac{AR^2}{3r^3}$.

Laplace: no max/min inside; $V(\mathbf{r})$ = average over any sphere around it. **Uniqueness:** V given on all boundaries (or Q on each conductor + V at ∞) ⇒ unique; anything that fits is THE answer.

Energy: $W = \frac{1}{2}\sum q_i V_i = \frac{1}{2}\int\rho V = \frac{\epsilon_0}{2}\int E^2$; pair $\frac{kq_1q_2}{r}$; shell $\frac{kQ^2}{2R}$, ball $\frac{3kQ^2}{5R}$; not additive (cross term $\epsilon_0\int\mathbf{E}_1\cdot\mathbf{E}_2$).

B2 Fields of standard charges

point	$E = \frac{kq}{r^2}$, $V = \frac{kq}{r}$
line λ	$\frac{\lambda}{2\pi\epsilon_0 s}$, $V = -\frac{\lambda}{2\pi\epsilon_0}\ln\frac{s}{s_0}$
sheet σ	$\frac{\sigma}{2\epsilon_0}$ each side; 2 opposite: $\frac{\sigma}{\epsilon_0}$ between
ring axis	$E = \frac{kQz}{(z^2+R^2)^{3/2}}$, $V = \frac{kQ}{\sqrt{z^2+R^2}}$
disk axis	$\frac{\sigma}{2\epsilon_0}(1 - \frac{z}{\sqrt{z^2+R^2}})$
shell	out $\frac{kQ}{r^2}$, in 0; $V_{in} = \frac{kQ}{R}$
ball ρ	in $\frac{\rho r}{3\epsilon_0}$; $V_{in} = \frac{kQ}{2R}(3 - \frac{r^2}{R^2})$
cyl ρ	in $\frac{\rho s}{2\epsilon_0}$; out $\frac{\rho a^2}{2\epsilon_0 s}$
slab ρ, |z|<a	in $\frac{\rho z}{\epsilon_0}$; out $\pm\frac{\rho a}{\epsilon_0}$ const
ball $\rho = kr^n$	$E = \frac{kr^{n+1}}{(n+3)\epsilon_0}$
cyl $\rho = ks^n$	$E = \frac{ks^{n+1}}{(n+2)\epsilon_0}$

B3 More Coulomb/Gauss results

segment λ	$E_\perp = \frac{k\lambda}{s}(\sin\theta_2 - \sin\theta_1)$, $E_\parallel = \frac{k\lambda}{s}(\cos\theta_2 - \cos\theta_1)$ (θ from ⊥ foot)
sym. segment ±L	$E = \frac{2k\lambda L}{s\sqrt{s^2+L^2}}$; $V = 2k\lambda\ln\frac{L+\sqrt{L^2+s^2}}{s}$
disk axis V	$\frac{\sigma}{2\epsilon_0}(\sqrt{z^2+R^2} - \|z\|)$
hemishell σ, center	$E = \frac{\sigma}{4\epsilon_0}$; solid hemisphere ρ: $\frac{\rho R}{4\epsilon_0}$
cyl shell σ	$\frac{\sigma R}{\epsilon_0 s}$ out, 0 in
slab ρ(z), symmetric	$E(z) = \frac{1}{\epsilon_0}\int_0^z \rho\,dz'$
conductor surface	$E = \frac{\sigma}{\epsilon_0}$; sphere $\sigma = \frac{Q}{4\pi R^2}$
grounded sphere + q at a	$\sigma = -\frac{q}{4\pi R}\frac{a^2-R^2}{(R^2+a^2-2Ra\cos\theta)^{3/2}}$
plane + q	bring in: $W = -\frac{kq^2}{4d}$

B4 Induced dipoles & cavities

Sphere (R) far from q (distance r≫R): $\mathbf{p} = 4\pi\epsilon_0 R^3\beta\mathbf{E}$, $\beta = \frac{\epsilon_r-1}{\epsilon_r+2}$ (conductor β=1); force $F = \frac{2kq^2R^3\beta}{r^5}$ attractive; $\alpha_{cond} = 4\pi\epsilon_0 R^3$.

Ball ρ with spherical cavity offset $\mathbf{d}$: $\mathbf{E}_{cav} = \frac{\rho\mathbf{d}}{3\epsilon_0}$ uniform. Cylinder ρ with cyl. cavity: $\frac{\rho\mathbf{d}}{2\epsilon_0}$. Slab ρ with cavity: superpose $\frac{\rho z}{\epsilon_0}\hat{z} - \frac{\rho(\mathbf{r}-\mathbf{c})}{3\epsilon_0}$.

$\mathbf{P} = k\hat{r}$ (n=0): $\rho_b = -\frac{2k}{r}$, $\sigma_b = k$, $\mathbf{E}_{in} = -\frac{k}{\epsilon_0}\hat{r}$. $\mathbf{P} = \frac{k}{r^2}\hat{r}$ in shell a..b: $\rho_b = 0$, $\sigma_b(a) = -\frac{k}{a^2}$, $\sigma_b(b) = \frac{k}{b^2}$, $E = -\frac{k}{\epsilon_0 r^2}$ in shell.

Electron cloud $\rho = -\frac{q}{\pi a^3}e^{-2r/a}$: $Q_{cloud}(r) = -q[1 - e^{-2r/a}(1 + \frac{2r}{a} + \frac{2r^2}{a^2})]$. $\int_0^\infty x^n e^{-x/a}dx = n!\,a^{n+1} \cdot \int_0^s xe^{-x/a}dx = a^2[1 - (1+\frac{s}{a})e^{-s/a}]$.

B5 Units

P, D, σ: C/m² · E: V/m · p: C·m · ρ: C/m³ · ϵ: F/m · χ_e, ϵ_r: none

B6 Constants

$e = 1.60\times10^{-19}$ C · $m_e = 9.11\times10^{-31}$ kg · $\epsilon_0 = 8.85\times10^{-12}$ · $\mu_0 = 1.257\times10^{-6}$ · $c = 3.00\times10^8$ · ϵ_r: air 1.0005, glass 4–7, Si 11.8, water 80.

C · Multipoles & dipoles

C1 Multipole expansion

$V = k\sum_n \frac{1}{r^{n+1}}\int r'^n P_n(\cos\alpha)\rho\,d\tau' = k[\frac{Q}{r} + \frac{\mathbf{p}\cdot\hat{r}}{r^2} + \frac{1}{r^3}\sum q_i r_i^2 P_2(\cos\alpha_i) + \ldots]$

$Q = \int\rho$, $\mathbf{p} = \int\mathbf{r}'\rho\,d\tau' = \sum q_i\mathbf{r}_i$; shift origin by $\mathbf{a}$: $\mathbf{p}' = \mathbf{p} - Q\mathbf{a}$ (Q=0 ⇒ p fixed).

n-th term: $V \propto r^{-(n+1)}$, $E \propto r^{-(n+2)}$; moment units C·mⁿ.

P_0=1, P_1=x, P_2=$\frac{3x^2-1}{2}$, P_3=$\frac{5x^3-3x}{2}$; $\int_{-1}^1 P_l P_{l'}dx = \frac{2\delta_{ll'}}{2l+1}$; $P_l(1) = 1$, $P_l(-1) = (-1)^l$.

Lin. quadrupole ($q, -2q, q$, spacing a): $V = \frac{kqa^2(3\cos^2\theta - 1)}{r^3}$. Square alternating: Q=p=0.

$\sigma = k_0\cos\theta$ shell: $p = \frac{4}{3}\pi R^3 k_0$; $V_{in} = \frac{k_0}{3\epsilon_0}r\cos\theta$, $V_{out} = \frac{k_0 R^3\cos\theta}{3\epsilon_0 r^2}$.

C2 More dipole moments

Hemishell σ (z>0): $p = \pi\sigma R^3$; ±σ hemishells: $2\pi\sigma R^3$. Solid hemisphere ρ: $\frac{\pi\rho R^4}{4}$; ±ρ halves: $\frac{\pi\rho R^4}{2}$. Shell $k\cos\theta$: $\frac{4}{3}\pi kR^3$. Ball $k\cos\theta$: $\frac{\pi kR^4}{3}$. Ring $\lambda_0\cos\phi$: $\pi R^2\lambda_0$. Rod $\lambda \propto z$: $\frac{2}{3}\lambda_0 L^2$. Square ±q alternating: p=0 (quadrupole). q at $\pm d/2$: qd.

C3 Dipole

$V = \frac{kp\cos\theta}{r^2} \cdot \mathbf{E} = \frac{kp}{r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta}) = \frac{k}{r^3}[3(\mathbf{p}\cdot\hat{r})\hat{r} - \mathbf{p}]$

axis $\frac{2kp}{z^3}$; equator $-\frac{kp}{r^3}$ (antiparallel)

$\mathbf{N} = \mathbf{p}\times\mathbf{E} \cdot U = -\mathbf{p}\cdot\mathbf{E} \cdot \mathbf{F} = (\mathbf{p}\cdot\nabla)\mathbf{E} = \nabla(\mathbf{p}\cdot\mathbf{E})$ · uniform E: F=0

2 dipoles: $U = \frac{k}{r^3}[\mathbf{p}_1\cdot\mathbf{p}_2 - 3(\mathbf{p}_1\cdot\hat{r})(\mathbf{p}_2\cdot\hat{r})]$; coaxial: $F = \frac{6kp_1p_2}{r^4} = \frac{3p_1p_2}{2\pi\epsilon_0 r^4}$ attract

Image of p in plane: ∥ flips, ⊥ kept; $N = \frac{kp^2\sin\theta\cos\theta}{(2z)^3}$, rests ⊥.

Induced: $\mathbf{p} = \alpha\mathbf{E}$, atom $\alpha = 4\pi\epsilon_0 a^3$; Clausius–Mossotti $\alpha = \frac{3\epsilon_0}{N}\frac{\epsilon_r-1}{\epsilon_r+2}$.

C4 Dipole extras

Dipole near charge q (aligned, distance r): $F = \frac{2kqp}{r^3}$ (attract); $U = -\frac{kqp}{r^2}$. Field energy of a dipole outside R: $\frac{p^2}{12\pi\epsilon_0 R^3}$. Uniform-P sphere: $\frac{\epsilon_0}{2}\int E^2 = \frac{2\pi R^3P^2}{9\epsilon_0}$ (⅓ in, ⅔ out).

Two charges q_1 at z_1, q_2 at z_2: $p = q_1z_1 + q_2z_2$; p vanishes about $z_0 = p/Q$. Rod $\lambda = \lambda_0 z/L$ on $[-L, L]$: $p = \frac{2}{3}\lambda_0 L^2$. Ball $\rho = k\cos\theta$: $p = \frac{\pi kR^4}{3}$. Ball $\rho = kz$: $p = \frac{4\pi kR^5}{15}$. Shell $\sigma = k\cos\theta$: $\frac{4}{3}\pi R^3 k$. Ring $\lambda = \lambda_0\cos\phi$: $p_x = \pi R^2\lambda_0$.

D · Polarization, D, linear dielectrics

D1 ★ D → E → P chain (always this order)

1. $\oint\mathbf{D}\cdot d\mathbf{a} = Q_{f,enc}$ (needs symmetry) · $\nabla\cdot\mathbf{D} = \rho_f$
2. $\mathbf{E} = \mathbf{D}/\epsilon$ (linear) · $\mathbf{E} = (\mathbf{D} - \mathbf{P})/\epsilon_0$ (frozen P)
3. $\mathbf{P} = \mathbf{D} - \epsilon_0\mathbf{E} = \epsilon_0\chi_e\mathbf{E} = \frac{\chi_e}{\epsilon_r}\mathbf{D} = (1 - \frac{1}{\epsilon_r})\mathbf{D}$
4. $\sigma_b = \mathbf{P}\cdot\hat{n}$ ($\hat{n}$ out of material; inner face $-\hat{r}$) · $\rho_b = -\nabla\cdot\mathbf{P} \cdot \sum q_b = 0$
5. $V = -\int\mathbf{E}\cdot d\mathbf{l}$ from ∞ inward, region by region

Linear homog.: $\rho_b = -\frac{\chi_e}{1+\chi_e}\rho_f$ · at conductor $\sigma_f + \sigma_b = \sigma_f/\epsilon_r$ · q_b at point q $= -\frac{\chi_e}{\epsilon_r}q$

No free + symmetry ⇒ D=0 ⇒ E=−P/ε₀ in, 0 out. Trap: $\nabla\times\mathbf{D} = \nabla\times\mathbf{P}$; no symmetry ⇒ D not from ρ_f alone. Free charge only on surface of dielectric ball ⇒ inside D=E=P=0.

D2 ★ Polarization P

$V = k\oint\frac{\sigma_b}{r}da' + k\int\frac{\rho_b}{r}d\tau' = k\int\frac{\mathbf{P}\cdot\hat{r}}{r^2}d\tau'$

Uniform: $\rho_b = 0$, $\sigma_b = \pm P$ on pierced faces. $\mathbf{E}_{in} = -N\mathbf{P}/\epsilon_0$: slab⊥ 1 · sphere ⅓ · cyl⊥ ½ · needle∥ 0.

Sphere: $\sigma_b = P\cos\theta$; $V_{in} = \frac{P}{3\epsilon_0}r\cos\theta$, $V_{out} = \frac{PR^3\cos\theta}{3\epsilon_0 r^2}$; $\mathbf{p} = \frac{4}{3}\pi R^3\mathbf{P}$.

Offset balls $\pm\rho$: overlap $\mathbf{E} = -\frac{\rho\mathbf{d}}{3\epsilon_0}$; cyls: $-\frac{\rho\mathbf{d}}{2\epsilon_0}$. Uniform-P shell: E=0 in cavity. Cavity in P-slab: $-\frac{2P}{3\epsilon_0}$.

$P = kr^n\hat{r}$: $\rho_b = -(n+2)kr^{n-1}$; $P = ks^n\hat{s}$: $-(n+1)ks^{n-1}$; $\sigma_b = P(R)$.

$\mathbf{P} = k\mathbf{r}$: ρ_b=$-3k$, σ_b=kR, $\mathbf{E}_{in}$=$-k\mathbf{r}/\epsilon_0$. $\frac{k}{r}\hat{r}$ shell: ρ_b=$-\frac{k}{r^2}$, $\sigma_b(a)$=$-\frac{k}{a}$, $\sigma_b(b)$=$\frac{k}{b}$, E=$-\frac{k}{\epsilon_0 r}$.

Bar electret: caps $\pm P$; long ≈ 2 point charges $\pm P\pi a^2$; short ≈ capacitor.

D3 Linear setups (free charge given)

q in ε: $E = \frac{q}{4\pi\epsilon r^2}$. Conductor Q (a) + shell εr to b: $D = \frac{Q}{4\pi r^2}$; $V(a) = kQ[\frac{1}{b} + \frac{1}{\epsilon_r}(\frac{1}{a} - \frac{1}{b})]$; $\sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi a^2}$; $\sigma_b(b) = +\frac{\chi_e}{\epsilon_r}\frac{Q}{4\pi b^2}$. Two shells: $\sigma_b(b) = \frac{Q}{4\pi b^2}(\frac{1}{\epsilon_{r2}} - \frac{1}{\epsilon_{r1}})$. Coated wire λ: $\sigma_b(a) = -\frac{\chi_e}{\epsilon_r}\frac{\lambda}{2\pi a}$.

Ball of εr, ρ_f: $D = \frac{\rho_f r}{3}$, $\sigma_b = \frac{\chi_e}{\epsilon_r}\frac{\rho_f R}{3}$. Ball $\rho_f = kr$: $D = \frac{kr^2}{4}$.

Frozen slab P in capacitor ±σ: $D = \sigma$; $E_{slab} = \frac{\sigma - P}{\epsilon_0}$; $\Delta V = \frac{\sigma d - Pt}{\epsilon_0}$.

Varying $\epsilon(x)$: $D = \sigma$ still, $\rho_b = -dP/dx \neq 0$.

D4 Point charge in dielectric ball

q at center of ball R (ϵ_r): $E_{in} = \frac{q}{4\pi\epsilon r^2}$, out $\frac{kq}{r^2}$; $q_b(0) = -\frac{\chi_e}{\epsilon_r}q$; $\sigma_b(R) = \frac{\chi_e}{\epsilon_r}\frac{q}{4\pi R^2}$; $V(R) = \frac{kq}{R}$.

E · Capacitors & energy

E1 Capacitance

plates	$\frac{\epsilon A}{d}$; stacked: $\frac{1}{C} = \sum\frac{d_i}{\epsilon_i A}$; side: $\sum\frac{\epsilon_i A_i}{d}$
sphere	$4\pi\epsilon_0 R$; shells $\frac{4\pi\epsilon ab}{b-a}$
coax/L	$\frac{2\pi\epsilon_0}{\sum\ln(b_i/a_i)/\epsilon_{ri}}$
half fill	stacked $\frac{2\epsilon_r}{1+\epsilon_r}C_0$; side $\frac{1+\epsilon_r}{2}C_0$
4.18	εr 2,1.5: $P = \frac{\sigma}{2}, \frac{\sigma}{3}$; $V = \frac{7\sigma a}{6\epsilon_0}$; σ_b mid $= \frac{\sigma}{6}$

E2 More capacitors

Half-fill by angle: sphere $C = \frac{2\pi\epsilon_0(1+\epsilon_r)ab}{b-a}$; coax $\frac{C}{L} = \frac{\pi\epsilon_0(1+\epsilon_r)}{\ln(b/a)}$. Conducting slab t: $\frac{\epsilon_0 A}{d-t}$. Dielectric slab t: $\frac{\epsilon_0 A}{d-t+t/\epsilon_r}$. ϵ linear $\epsilon_1 \to \epsilon_2$ across gap: $C = \frac{A(\epsilon_2-\epsilon_1)}{d\ln(\epsilon_2/\epsilon_1)}$. $\epsilon = \epsilon_0\frac{b}{r}$ sphere: $\frac{4\pi\epsilon_0 b}{\ln(b/a)}$.

E3 Energy & force

$W = \frac{\epsilon_0}{2}\int E^2 = \frac{1}{2}\int\rho V = \frac{1}{2}\sum q_i V_i$ (all charge) · dielectric: $W = \frac{1}{2}\int\mathbf{D}\cdot\mathbf{E} = \frac{1}{2}\int\frac{D^2}{\epsilon}$ (free work); frozen P, no free: $\frac{1}{2}\int D\cdot E = 0$.

$\frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$. Outside sphere: $\frac{Q^2}{8\pi\epsilon_0 R}$. Ball $\frac{3Q^2}{20\pi\epsilon_0 R}$ (in part $\frac{Q^2}{40\pi\epsilon_0 R}$); ball of εr: $\frac{Q^2}{8\pi\epsilon_0 R}(1 + \frac{1}{5\epsilon_r})$; 4.26: $\frac{Q^2}{8\pi\epsilon_0\epsilon_r}(\frac{1}{a} + \frac{\chi_e}{b})$; coax/L $\frac{\lambda^2}{4\pi\epsilon}\ln\frac{b}{a}$.

Fill: fixed Q → W÷εr; fixed V → W×εr. $F = \frac{1}{2}V^2\frac{dC}{dx}$ (toward ↑C): slab $\frac{\epsilon_0\chi_e wV^2}{2d}$; coax $\frac{\pi\epsilon_0\chi_e V^2}{\ln(b/a)}$, rise $h = \frac{\epsilon_0\chi_e V^2}{\rho g(b^2-a^2)\ln(b/a)}$.

Conductor surface: pressure $\frac{\sigma^2}{2\epsilon_0}$ outward. Infinite line: W/L diverges.

F · Boundary conditions & BVPs

F1 Boundary conditions

$D_1^\perp - D_2^\perp = \sigma_f \cdot \mathbf{E}^\parallel$ cont. · V cont. · $E^\perp$ jumps $\sigma_{tot}/\epsilon_0 \cdot \sigma_b = \mathbf{P}_1\cdot\hat{n}_1 + \mathbf{P}_2\cdot\hat{n}_2$

$\epsilon_1\partial_n V_1 - \epsilon_2\partial_n V_2 = -\sigma_f \cdot \frac{\tan\theta_1}{\tan\theta_2} = \frac{\epsilon_1}{\epsilon_2}$ · conductor: $\sigma = -\epsilon\partial_n V$

F2 Laplace solutions

Sph: $V = \sum(A_l r^l + \frac{B_l}{r^{l+1}})P_l(\cos\theta)$. In: $B_l = 0$; out: $A_l = 0$ (except $-E_0 r\cos\theta$).

Cyl (2D): $A_0 + B_0\ln s + \sum(A_n s^n + B_n s^{-n})(C\cos n\phi + D\sin n\phi)$.

Cart: $e^{\pm ky}\sin kx$; sinh, cosh. Coef: $\frac{2l+1}{2}\int V_0 P_l\sin\theta d\theta$.

F3 ★ Objects in uniform $E_0\hat{z}$

Dielectric sphere: $V_{in} = -\frac{3E_0}{\epsilon_r+2}r\cos\theta$; $V_{out} = -E_0 r\cos\theta + \frac{\epsilon_r-1}{\epsilon_r+2}\frac{E_0R^3\cos\theta}{r^2}$

$\mathbf{P} = 3\epsilon_0\frac{\epsilon_r-1}{\epsilon_r+2}\mathbf{E}_0$; $\sigma_b = P\cos\theta$; $p = 4\pi\epsilon_0 R^3\frac{\epsilon_r-1}{\epsilon_r+2}E_0$; $E_r^{out}(R, 0) = \frac{3\epsilon_r}{\epsilon_r+2}E_0$

Conducting sphere: $V = -E_0(r - \frac{R^3}{r^2})\cos\theta$; $\sigma = 3\epsilon_0 E_0\cos\theta$; $p = 4\pi\epsilon_0 R^3 E_0$. +charge Q: add $\frac{kQ}{r}$.

Conducting cyl⊥: $V = -E_0(s - \frac{R^2}{s})\cos\phi$, $\sigma = 2\epsilon_0 E_0\cos\phi$. Dielectric cyl⊥: $E_{in} = \frac{2}{\epsilon_r+1}E_0$.

General: sphere $\frac{3\epsilon_{out}}{\epsilon_{in}+2\epsilon_{out}}$; cyl $\frac{2\epsilon_{out}}{\epsilon_{in}+\epsilon_{out}}$; cavity $\frac{3\epsilon_r}{1+2\epsilon_r}$; $E_{in} = \frac{E_0}{1+N\chi_e}$.

Coated conductor (4.24): $V = A(r - \frac{a^3}{r^2})\cos\theta$, $A = \frac{-3E_0}{(\epsilon_r+2)+2(\epsilon_r-1)a^3/b^3}$.

Check: εr→1 vacuum; εr→∞ conductor.

F4 Image BC equations (write these)

q above εr: V cont. $q + q' = q''$; $\epsilon\partial_z V$ cont. $q - q' = \epsilon_r q''$. Sphere in E_0: $AR = -E_0R + \frac{B}{R^2}$; $\epsilon_r A = -E_0 - \frac{2B}{R^3}$. Shell σ: V cont., $\partial_r V_{out} - \partial_r V_{in} = -\frac{\sigma}{\epsilon_0}$ at R.

F5 2D (cylinder) results

Line dipole (p per length): $V = \frac{p\cos\phi}{2\pi\epsilon_0 s}$. Uniform P ⊥ axis: $\sigma_b = P\cos\phi$, $\mathbf{E}_{in} = -\frac{\mathbf{P}}{2\epsilon_0}$. Dielectric cyl in $E_0\hat{x}$: $V_{in} = -\frac{2E_0}{\epsilon_r+1}s\cos\phi$, $V_{out} = -E_0 s\cos\phi + \frac{\epsilon_r-1}{\epsilon_r+1}\frac{E_0R^2\cos\phi}{s}$. Coax half-filled by angle ($\epsilon_1|\epsilon_2$): E shared, $E = \frac{\lambda}{\pi s(\epsilon_1+\epsilon_2)}$.

F6 2D polar shells

Circle at $V_0(\phi)$: in $a_0 + \sum(\frac{s}{R})^n(a_n\cos n\phi + b_n\sin n\phi)$, out same with $(\frac{R}{s})^n$ (+ $\ln s$ term if net λ).

Cyl shell $\sigma_0\cos\phi$: $V_{in} = \frac{\sigma_0}{2\epsilon_0}s\cos\phi$, $V_{out} = \frac{\sigma_0 R^2\cos\phi}{2\epsilon_0 s}$; $\mathbf{E}_{in} = -\frac{\sigma_0}{2\epsilon_0}\hat{x}$ uniform. Shell $\sigma_n\cos n\phi$: $V_{in} = \frac{\sigma_n s^n\cos n\phi}{2n\epsilon_0 R^{n-1}}$, $V_{out} = V_{in}\cdot(\frac{R}{s})^{2n}$.

F7 Separation of variables (Cartesian)

Slot V_0 on end: $V = \frac{4V_0}{\pi}\sum_{odd}\frac{1}{n}e^{-n\pi x/a}\sin\frac{n\pi y}{a}$; $c_n = \frac{2}{a}\int_0^a V_0(y)\sin\frac{n\pi y}{a}dy$; const V_0: $\frac{4V_0}{n\pi}$ odd n. Finite box: sinh/cosh.

G · Images

G1 Images

Grounded plane: $-q$ at mirror; $F = \frac{kq^2}{(2d)^2}$; $\sigma = \frac{-qd}{2\pi(s^2+d^2)^{3/2}}$; $W = -\frac{kq^2}{4d}$. Plane at V_0: add V_0 .

Grounded sphere R, q at a: $q' = -\frac{R}{a}q$ at $\frac{R^2}{a}$; $F = \frac{kq^2Ra}{(a^2-R^2)^2}$. Neutral: add $+\frac{R}{a}q$ at center. At V_0: add $4\pi\epsilon_0 RV_0$ at center.

Dielectric half-space: above $q' = -\frac{\epsilon_r-1}{\epsilon_r+1}q$ at $-d$; below $q'' = \frac{2q}{\epsilon_r+1}$ at $+d$ (use k). $\sigma_b = -\frac{1}{2\pi}\frac{\chi_e}{\chi_e+2}\frac{qd}{(s^2+d^2)^{3/2}}$; $F = \frac{kqq'}{4d^2}$; W = ½ interaction.

Line λ, plane: $-\lambda$ (× $\frac{\epsilon_r-1}{\epsilon_r+1}$ for dielectric). Corner 90°: 3 images.

H · Math

H1 Vector calculus (cyl)

$\nabla f = f_s\hat{s} + \frac{1}{s}f_\phi\hat{\phi} + f_z\hat{z} \cdot \nabla\cdot\mathbf{A} = \frac{1}{s}\partial_s(sA_s) + \frac{1}{s}\partial_\phi A_\phi + \partial_z A_z$

$\nabla\times\mathbf{A} = [\frac{1}{s}\partial_\phi A_z - \partial_z A_\phi]\hat{s} + [\partial_z A_s - \partial_s A_z]\hat{\phi} + \frac{1}{s}[\partial_s(sA_\phi) - \partial_\phi A_s]\hat{z}$

$\nabla^2 f = \frac{1}{s}\partial_s(sf_s) + \frac{1}{s^2}f_{\phi\phi} + f_{zz}$ · sph: $\nabla\cdot(F\hat{r}) = \frac{1}{r^2}(r^2F)'$

$\nabla\times\nabla f = 0$, $\nabla\cdot\nabla\times\mathbf{A} = 0$, $\nabla\times\nabla\times\mathbf{A} = \nabla(\nabla\cdot\mathbf{A}) - \nabla^2\mathbf{A}$; $\nabla\cdot(f\mathbf{A}) = f\nabla\cdot\mathbf{A} + \mathbf{A}\cdot\nabla f$.

$\nabla\cdot\frac{\hat{r}}{r^2} = 4\pi\delta^3$; $\nabla\cdot\frac{\hat{s}}{s} = 2\pi\delta^2$; $\nabla^2\frac{1}{r} = -4\pi\delta^3$.

H2 Spherical calculus

$\nabla f = f_r\hat{r} + \frac{1}{r}f_\theta\hat{\theta} + \frac{1}{r\sin\theta}f_\phi\hat{\phi} \cdot \nabla\cdot\mathbf{A} = \frac{1}{r^2}\partial_r(r^2A_r) + \frac{\partial_\theta(\sin\theta A_\theta)}{r\sin\theta} + \frac{\partial_\phi A_\phi}{r\sin\theta}$

$\nabla\times\mathbf{A} = \frac{\hat{r}}{r\sin\theta}[\partial_\theta(\sin\theta A_\phi) - \partial_\phi A_\theta] + \frac{\hat{\theta}}{r}[\frac{\partial_\phi A_r}{\sin\theta} - \partial_r(rA_\phi)] + \frac{\hat{\phi}}{r}[\partial_r(rA_\theta) - \partial_\theta A_r]$

$\nabla^2 f = \frac{1}{r^2}\partial_r(r^2 f_r) + \frac{\partial_\theta(\sin\theta f_\theta)}{r^2\sin\theta} + \frac{f_{\phi\phi}}{r^2\sin^2\theta} \cdot d\tau = r^2\sin\theta\,dr\,d\theta\,d\phi \cdot da_{sph} = R^2\sin\theta\,d\theta\,d\phi$

H3 Unit vectors & cross products

$\hat{r} = \sin\theta\cos\phi\hat{x} + \sin\theta\sin\phi\hat{y} + \cos\theta\hat{z} \cdot \hat{\theta} = \cos\theta\cos\phi\hat{x} + \cos\theta\sin\phi\hat{y} - \sin\theta\hat{z} \cdot \hat{\phi} = -\sin\phi\hat{x} + \cos\phi\hat{y}$

$\hat{z} = \cos\theta\hat{r} - \sin\theta\hat{\theta} \cdot \hat{s} = \cos\phi\hat{x} + \sin\phi\hat{y} \cdot \hat{s} = \sin\theta\hat{r} + \cos\theta\hat{\theta}$ · equator: $\hat{\theta} = -\hat{z}$

$\hat{x}\times\hat{y} = \hat{z}$, $\hat{y}\times\hat{z} = \hat{x}$, $\hat{z}\times\hat{x} = \hat{y} \cdot \hat{s}\times\hat{\phi} = \hat{z}$, $\hat{\phi}\times\hat{z} = \hat{s}$, $\hat{z}\times\hat{s} = \hat{\phi} \cdot \hat{r}\times\hat{\theta} = \hat{\phi}$, $\hat{\theta}\times\hat{\phi} = \hat{r}$, $\hat{\phi}\times\hat{r} = \hat{\theta}$

H4 Legendre & series extras

$P_4 = \frac{35x^4-30x^2+3}{8}$; $x^2 = \frac{1}{3}P_0 + \frac{2}{3}P_2$; $x^3 = \frac{3}{5}P_1 + \frac{2}{5}P_3$; $x^4 = \frac{1}{5}P_0 + \frac{4}{7}P_2 + \frac{8}{35}P_4$.

$P_l(0)$: $P_2 = -\frac{1}{2}$, $P_4 = \frac{3}{8}$, odd 0. $\int_0^1 P_l$: $1, \frac{1}{2}, 0, -\frac{1}{8}$ (l=0..3).

Hemispheres $\pm V_0$: $c_1 = \frac{3}{2}V_0$, $c_3 = -\frac{7}{8}V_0$ (odd only).

$\frac{1}{|\mathbf{r}-\mathbf{r}'|} = \sum\frac{r_<^l}{r_>^{l+1}}P_l(\cos\alpha) \cdot \sum_{odd}\frac{1}{n^2} = \frac{\pi^2}{8} \cdot \sum\frac{1}{n^2} = \frac{\pi^2}{6} \cdot \sum_{odd}\frac{(-1)^{(n-1)/2}}{n} = \frac{\pi}{4}$

H5 Integrals

$\int\frac{dz}{(z^2+s^2)^{3/2}} = \frac{z}{s^2\sqrt{z^2+s^2}} \cdot \int\frac{z\,dz}{(z^2+s^2)^{3/2}} = \frac{-1}{\sqrt{z^2+s^2}} \cdot \int\frac{dz}{\sqrt{z^2+s^2}} = \ln(z + \sqrt{z^2+s^2}) \cdot \int\frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$

$\int_0^\pi\sin\theta = 2$, $\int_0^\pi\sin^3\theta = \frac{4}{3}$, $\int_0^\pi\cos^2\theta\sin\theta = \frac{2}{3}$, $\int_0^{2\pi}\cos^2\phi = \pi$. Partial fractions: $\frac{a}{r^2(r+a)} = \frac{1}{r^2} - \frac{1}{ar} + \frac{1}{a(r+a)}$.

I · HW answers

I1 HW answers: dielectrics

4.6 image $\mathbf{p}' = (-p_x, -p_y, p_z)$ at −z; $N = \frac{p^2\sin\theta\cos\theta}{4\pi\epsilon_0(2z)^3}$; rests θ=0,π. **4.7** $U = -\mathbf{p}\cdot\mathbf{E}$ (work $\int qE$ over d). **4.10** σ_b=kR, ρ_b=$-3k$, E_{in}=$-\frac{kr}{\epsilon_0}$, out 0. **4.11** caps ±P, sides 0. **4.12** $V = \frac{\mathbf{P}}{\rho} \cdot \mathbf{E}_{ball}$: in $\frac{Pr\cos\theta}{3\epsilon_0}$, out $\frac{PR^3\cos\theta}{3\epsilon_0 r^2}$. **4.15** ρ_b=$-\frac{k}{r^2}$, $\sigma_b(b)$=$\frac{k}{b}$, $\sigma_b(a)$=$-\frac{k}{a}$, E=$-\frac{k}{\epsilon_0 r}$ in shell only. **4.18** D=σ; E=σ/2ε₀, 2σ/3ε₀; P=σ/2, σ/3; V=7σa/6ε₀; σ_b: −σ/2 top, +σ/2|−σ/3 mid, +σ/3 bottom. **4.19** C/C_0: (a) $\frac{2\epsilon_r}{1+\epsilon_r}$ (b) $\frac{1+\epsilon_r}{2}$. (a) D same, $E_{air} = \frac{2\epsilon_r V}{(1+\epsilon_r)d}$, $E_{diel} = \frac{E_{air}}{\epsilon_r}$; (b) $E = \frac{V}{d}$ both, $\sigma_f = \epsilon E$. **4.21** $\frac{C}{L} = \frac{2\pi\epsilon_0}{\ln\frac{b}{a} + \frac{1}{\epsilon_r}\ln\frac{c}{b}}$. **4.24** $E = -A[(1 + \frac{2a^3}{r^3})\cos\theta\hat{r} - (1 - \frac{a^3}{r^3})\sin\theta\hat{\theta}]$. **4.26** $W = \frac{Q^2}{8\pi\epsilon_0(1+\chi_e)}(\frac{1}{a} + \frac{\chi_e}{b})$.

PHYS 435 · Exam II · BACK · J how-to · K Biot–Savart · L Ampère · M A · N dipoles · O matter · P BCs · Q Exam I · R answers

$\mu_0 = 4\pi \times 10^{-7}$ T·m/A · $\frac{\mu_0}{4\pi} = 10^{-7}$ · $c = 1/\sqrt{\mu_0\epsilon_0}$ · RHR: thumb I, fingers B

FIND IT FAST: problem has… → box

wire bent into lines/arcs	**J2 K2 K3**
wire, cylinder, sheet, slab, solenoid, toroid	**J3 L1 L2**
cylinder with off-axis hole	**L4**
find A (vector potential)	**M1 M2**
given B, A or M: find J	**M3 J4**
spinning charge, m, torque, force	**N1 N2 N3**
frozen M; linear µ material	**J4 O1 O3**
point off the line near wires	**K4**
BCs at a current sheet	**P1**
E ↔ B analogies	**P2 P3**
Exam I separation / images	**Q1 Q2**
HW / mock answers	**R1 R2**

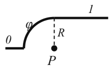
★ = most used. Codes: letter = topic group, number = box.

J · HOW TO SOLVE · magnetic

J1 ★ Which method? (magnetic)

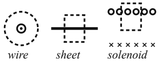
wire bent into lines/arcs	Biot–Savart pieces
wire, cyl., coax, sheet, slab, solenoid, toroid	Ampère
hole / overlap	superpose ±J full shapes
B, A or M given	curl/div → J, J_b, K_b
need A	A ∥ J; one curl component ODE / Stokes
spinning charge	K=σv, J=ρv; $m = \frac{Q}{2M}L$
small loop / far away	dipole $\mathbf{m} = I\mathbf{a}$
material + free I	H by Ampère → B=µH
frozen M	J_b, K_b; or H=0 by symmetry
force/torque on loop	$\oint I d\mathbf{l} \times \mathbf{B}$; uniform: F=0, N=m×B

J2 ★ Recipe: Biot–Savart by pieces



1. Chop into straight pieces + arcs.
2. Cross out every piece whose **line** passes through P (0).
3. Segment $\frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$; semi-∞ beside end $\frac{\mu_0 I}{4\pi s}$ ("1"); arc $\frac{\mu_0 I\phi}{4\pi R}$.
4. Direction each by $d\mathbf{l} \times \hat{r}$ (RHR); add with signs.
5. Not in a plane (axis points): write $d\mathbf{B}$, keep the component symmetry leaves (loop axis: $\times\frac{R}{\sqrt{R^2+z^2}}$), integrate.

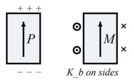
J3 ★ Recipe: Ampère



1. **Symmetry sentence:** $B_s = 0$ (∇·B=0), $B_z = 0$ (rectangle to ∞) ⇒ $\mathbf{B} = B(s)\hat{\phi}$. Sheet: B ∥ sheet ⊥ K. Solenoid: B ∥ axis.
2. **Loop:** circle (wire) · rectangle straddling (sheet/slab) · rectangle half inside (solenoid) · circle in core (toroid).
3. $I_{enc} = \int \mathbf{J}\cdot d\mathbf{a}$ ($\int J(s)2\pi s\, ds$), or KL, or nIL.
4. Solve $B\,2\pi s = \mu_0 I_{enc}$ · $2BL = \mu_0 KL$ · $BL = \mu_0 nIL$.
5. Direction by RHR; B cont. unless K (jump µ₀K).

Matter: same with H and I_f, then $B = \mu H$.

J4 ★ Recipe: magnetized object / H



1. $\mathbf{J}_b = \nabla\times\mathbf{M}$ (one-liner); $\mathbf{K}_b = \mathbf{M}\times\hat{n}$ (n̂ out; faces ∥ M).
2. Check net bound current = 0.
3. **Shortcut:** no free I + symmetry ⇒ H=0 ⇒ B=µ₀M.
4. Else recognize: sphere sinθ φ̂ ⇒ spinning shell; side current ⇒ solenoid; or Ampère with $J_b + K_b$.
5. Free I present: H from I_f, $\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M})$ or µH.

P vs M: P charges the caps, M puts current on the sides.

K · Currents & Biot–Savart

K1 Currents & force

$\mathbf{F} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$; cyclotron $r = \frac{mv}{qB}$, $\omega = \frac{qB}{m}$; B does no work.

$I = \lambda v$, $\mathbf{K} = \sigma\mathbf{v}$ (A/m), $\mathbf{J} = \rho\mathbf{v}$ (A/m²); $I = \int \mathbf{J}\cdot d\mathbf{a}$. $\nabla\cdot\mathbf{J} = -\partial_t\rho$; statics $\nabla\cdot\mathbf{J} = 0$.

Spin: disk $K = \sigma\omega s$; shell $K = \sigma\omega R\sin\theta$; ball $J = \rho\omega r\sin\theta$ ($\hat{\phi}$). Ring $I = \frac{Q\omega}{2\pi}$.

$\mathbf{F} = \int I\, d\mathbf{l}\times\mathbf{B} = \int \mathbf{K}\times\mathbf{B}\, da = \int \mathbf{J}\times\mathbf{B}\, d\tau$.

Uniform B: loop F=0; curved wire = chord. Wires: $f = \frac{\mu_0 I_1 I_2}{2\pi d}$ (same → attract). Sheet: $\mathbf{f} = \mathbf{K}\times\mathbf{B}_{avg}$; solenoid wall pressure $\frac{B^2}{2\mu_0}$.

Loop near wire: $\frac{\mu_0 I^2 a^2}{2\pi s(s+a)}$ toward. $\mathbf{B} = kz\hat{x}$, loop in yz: $Ika^2\hat{z}$.

K2 ★ Biot–Savart results

$\mathbf{B} = \frac{\mu_0}{4\pi}\int \frac{I d\mathbf{l}\times\hat{r}}{r^2}$. On a piece's line: 0.

segment	$\frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$
∞ wire	$\frac{\mu_0 I}{2\pi s}$; semi-∞ beside end $\frac{\mu_0 I}{4\pi s}$
arc φ	$\frac{\mu_0 I\phi}{4\pi R}$; loop $\frac{\mu_0 I}{2R}$; half $\frac{\mu_0 I}{4R}$
loop axis	$\frac{\mu_0 I R^2}{2(R^2+z^2)^{3/2}} \to \frac{\mu_0 I R^2}{2z^3}$
square (½side R)	$\frac{\sqrt{2}\mu_0 I}{\pi R}$; n-gon $\frac{n\mu_0 I}{2\pi R}\sin\frac{\pi}{n}$
rect $L_1\times L_2$	$\frac{2\mu_0 I\sqrt{L_1^2+L_2^2}}{\pi L_1 L_2}$
square axis	$\frac{\mu_0 I a^2}{2\pi(z^2+a^2/4)\sqrt{z^2+a^2/2}}$
finite sol.	$\frac{\mu_0 nI}{2}(\cos\theta_2 - \cos\theta_1)$; mouth $\frac{\mu_0 nI}{2}$
Helmholtz	mid $\frac{\mu_0 I R^2}{(R^2+d^2/4)^{3/2}}$; d=R: $\frac{8\mu_0 I}{5\sqrt{5}R}$
hairpin	$\frac{\mu_0 I}{4R} + \frac{\mu_0 I}{2\pi R}$
spin disk axis	$\frac{\mu_0\sigma\omega}{2}[\frac{R^2+2z^2}{\sqrt{R^2+z^2}} - 2z]$; center $\frac{\mu_0\sigma\omega R}{2}$

K3 More loops

D-shape (semicircle + diameter) center: $\frac{\mu_0 I}{4R}$. Two concentric arcs φ (a,b) + radial: $\frac{\mu_0 I\phi}{4\pi}(\frac{1}{a} - \frac{1}{b})$; m = $\frac{I\phi}{2}(b^2 - a^2)$. Loop tangent leads: $\frac{\mu_0 I}{4\pi R}(\phi + n_{tan})$. Semicircle in uniform $\mathbf{B}$: $\mathbf{F} = I(2R\hat{x})\times\mathbf{B}$. Three sides of square (2R): $\frac{\sqrt{5}\mu_0 I}{4\pi R}$.

K4 Any straight wire, any point

Wire ∥ z through (x_0, y_0), current I: $\mathbf{B} = \frac{\mu_0 I}{2\pi}\frac{(-(y-y_0), x-x_0, 0)}{(x-x_0)^2+(y-y_0)^2}$. Wire ∥ y through $(x_0, \cdot, z_0)$: $\frac{\mu_0 I}{2\pi}\frac{(z-z_0, 0, -(x-x_0))}{(x-x_0)^2+(z-z_0)^2}$.

5.67: $\mathbf{B} = \frac{\mu_0 I}{2\pi}\left[\frac{(z,0,a-x)}{(x-a)^2+z^2} + \frac{(-y,x+a,0)}{(x+a)^2+y^2}\right] + \mathbf{B}_0$.

Flux of wire through coplanar rect. (width a from s, height h): $\Phi = \frac{\mu_0 Ih}{2\pi}\ln\frac{s+a}{s}$. Through toroid turn: $\frac{\mu_0 NIh}{2\pi}\ln\frac{b}{a}$.

Segment, angles α from the wire line: $\frac{\mu_0 I}{4\pi s}|\cos\alpha_1 - \cos\alpha_2|$. Point on extension: 0.

Work to flip m in B: $2mB$. Dipole in field of wire: $\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B})$.

L · Ampère & superposition

L1 ★ Ampère results

$\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 I_{enc}$ · $\nabla\times\mathbf{B} = \mu_0\mathbf{J}$ · $\nabla\cdot\mathbf{B} = 0$

wire uniform	in $\frac{\mu_0 Is}{2\pi a^2} = \frac{\mu_0 Js}{2}$; out $\frac{\mu_0 I}{2\pi s}$
$J \propto s^k$	$I_{enc} = I(\frac{s}{a})^{k+2}$, $B = \frac{\mu_0 I}{2\pi s}(\frac{s}{a})^{k+2}$
surface I	0 in; $\frac{\mu_0 I}{2\pi s}$ out
tube a..b	$\frac{\mu_0 I}{2\pi s}\frac{s^2-a^2}{b^2-a^2}$
coax	$\frac{\mu_0 I}{2\pi s}$ between, 0 out
sheet K	$\frac{\mu_0 K}{2}$, dir $\mathbf{K}\times\hat{n}$; ± sheets: $\mu_0 K$ / 0
slab $J\hat{x}$	$-\mu_0 Jz\hat{y}$ in; $\mp\mu_0 Ja\hat{y}$ out
solenoid	$\mu_0 nI = \mu_0 K$ in, 0 out
toroid	$\frac{\mu_0 NI}{2\pi s}$; flux (rect) $\frac{\mu_0 NIh}{2\pi}\ln\frac{b}{a}$
2 sols opp.	$\mu_0(n_1 - n_2)I$, $\mu_0 n_2 I$, 0
moving cap.	$\mu_0\sigma v$; $f_B = \frac{\mu_0\sigma^2 v^2}{2}$ = f_E at $v = c$
$\mathbf{J} = \frac{k}{s}\hat{z}$ tube	$\frac{\mu_0 k(s-a)}{s}$

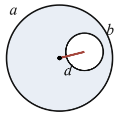
L2 More B setups

tube, axial K	0 in; $\frac{\mu_0 KR}{s}\hat{\phi}$ out; $A_z = -\mu_0 KR\ln\frac{s}{R}$ out
toroid circ. section	$\Phi = \mu_0 NI(b - \sqrt{b^2 - a^2})$
M-cyl center (L, a)	$\frac{\mu_0 ML}{\sqrt{L^2+4a^2}}$
spinning ring axis	loop formula with $I = \frac{Q\omega}{2\pi}$
two ⊥ sheets	add vectors, each $\frac{\mu_0 K}{2}$
helix	$r = \frac{mv_\perp}{qB}$, pitch $\frac{2\pi mv_\parallel}{qB}$
velocity selector	$v = E/B$
solenoid + core χ_m	$B = \mu nI$; $H = nI$; $M = \chi_m nI$
core in part of cross-section	H same $= nI$; B: μnI | $\mu_0 nI$

L3 More B results

spin cyl shell σ	$K = \sigma\omega R \Rightarrow B = \mu_0\sigma\omega R$ in, 0 out
spin solid cyl ρ	$J = \rho\omega s \Rightarrow B(s) = \frac{\mu_0\rho\omega}{2}(R^2 - s^2)$
spin ball center	$\frac{\mu_0\rho\omega R^2}{3} = \frac{\mu_0 Q\omega}{4\pi R}$
M-rod length L, axis	$\frac{\mu_0 M}{2}(\cos\theta_2 - \cos\theta_1)$; end face $\frac{\mu_0 M}{2}$
M-sphere + $\mathbf{B}_0$	in $\mathbf{B}_0 + \frac{2}{3}\mu_0\mathbf{M}$; $\mathbf{H} = \frac{\mathbf{B}_0}{\mu_0} - \frac{\mathbf{M}}{3}$
sheet + $B_0\hat{z}$	$\mathbf{f} = \mathbf{K}\times\mathbf{B}_0$ (own field averages 0)
E⊥B crossed	drift $v = E/B$; cycloid $R = \frac{mE}{qB^2}$
loop in **B**	$N = mB\sin\theta$; $U = -mB\cos\theta$
wire ∥ sheet K	$f/L = I\frac{\mu_0 K}{2}$ toward (∥ currents)

L4 ★ Off-axis hole (superposition)



$J = \frac{I}{\pi(a^2-b^2)}$ · $I_+ = \frac{Ia^2}{a^2-b^2}$ · $I_- = -\frac{Ib^2}{a^2-b^2}$

Uniform wire: in $\frac{\mu_0}{2}\mathbf{J}\times\mathbf{s}$, out $\frac{\mu_0 I}{2\pi s}\hat{\phi}$ (own axis).

Hole: $\mathbf{B} = \frac{\mu_0}{2}\mathbf{J}\times\mathbf{d}$, $|B| = \frac{\mu_0 Id}{2\pi(a^2-b^2)}$ uniform ⊥ d.

Metal: $\frac{\mu_0}{2}\mathbf{J}\times\mathbf{s} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$. Out: $\frac{\mu_0 I_+}{2\pi s}\hat{\phi} + \frac{\mu_0 I_-}{2\pi s'}\hat{\phi}'$.

Line through axes, at $(x, 0)$, x>a: $B_y = \frac{\mu_0}{2\pi}[\frac{I_+}{x} + \frac{I_-}{x-d}]$. At $(-x_0, 0)$ in metal: $\frac{\mu_0 J}{2}(\frac{b^2}{d+x_0} - x_0)$.

Opposite-J overlap: $\frac{\mu_0}{2}\mathbf{J}\times(\mathbf{c}_- - \mathbf{c}_+)$. $A_z^{hole} = -\frac{\mu_0 J}{4}(s^2 - |\mathbf{s} - \mathbf{d}|^2)$.

Slab $J\hat{x}$ w/ hole along x: $\mathbf{B} = -\frac{\mu_0 J}{2}(z\hat{y} + y\hat{z})$.

M · Vector potential

M1 ★ Vector potential A

$\mathbf{B} = \nabla\times\mathbf{A}$; Coulomb $\nabla\cdot\mathbf{A} = 0$; $\nabla^2\mathbf{A} = -\mu_0\mathbf{J}$; $\mathbf{A} = \frac{\mu_0}{4\pi}\int\frac{\mathbf{J}}{r}d\tau'$ (local currents only); $\oint \mathbf{A}\cdot d\mathbf{l} = \Phi_B$; gauge $\mathbf{A} + \nabla\lambda$.

No-integral recipe: A ∥ J → one curl component → ODE → A continuous, regular on axis.

$A_z(s)$: $B_\phi = -A_z'$ · $A_\phi(s)$: $B_z = \frac{1}{s}(sA_\phi)'$ · $A_x(z)$: $B_y = A_x'$ · $A_\phi(r,\theta)$: $B_r = \frac{\partial_\theta(\sin\theta A_\phi)}{r\sin\theta}$, $B_\theta = -\frac{\partial_r(rA_\phi)}{r}$

uniform B	$\frac{1}{2}\mathbf{B}\times\mathbf{r} = \frac{B_0 s}{2}\hat{\phi}$
wire	$-\frac{\mu_0 I}{2\pi}\ln\frac{s}{s_0}\hat{z}$
uniform J wire	in $\frac{\mu_0 J}{4}(a^2 - s^2)$; out $-\frac{\mu_0 I}{2\pi}\ln\frac{s}{a}$
$J = J_0\frac{s}{a}$	in $\frac{\mu_0 J_0(a^3-s^3)}{9a}$
solenoid	$\frac{\mu_0 nIs}{2}$ in; $\frac{\mu_0 nIR^2}{2s}$ out ($\hat{\phi}$)
sheet $K\hat{x}$	$-\frac{\mu_0 K}{2}\|z\|\hat{x}$
slab $J\hat{x}$	in $-\frac{\mu_0 Jz^2}{2}$; out $-\mu_0 Ja\|z\| + \frac{\mu_0 Ja^2}{2}$
spin shell	in $\frac{\mu_0 R\omega\sigma}{3}r\sin\theta$; out $\frac{\mu_0 R^4\omega\sigma}{3}\frac{\sin\theta}{r^2}$
dipole	$\frac{\mu_0}{4\pi}\frac{\mathbf{m}\times\hat{r}}{r^2} = \frac{\mu_0 m\sin\theta}{4\pi r^2}\hat{\phi}$
finite wire	$\frac{\mu_0 I}{4\pi}\ln\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}$
$A = k\hat{\phi}$	$B = \frac{k}{s}\hat{z}$, $J = \frac{k}{\mu_0 s^2}\hat{\phi}$
$A = ks^3\hat{z}$	$B = -3ks^2\hat{\phi}$, $J = -\frac{9ks}{\mu_0}\hat{z}$

Multipole: $\mathbf{A} = \frac{\mu_0 I}{4\pi}\sum\frac{1}{r^{n+1}}\oint r'^n P_n(\cos\alpha)d\mathbf{l}'$; monopole term = 0.

M2 More A results

M-sphere	in $\frac{\mu_0 M}{3}r\sin\theta\hat{\phi}$; out $\frac{\mu_0 MR^3}{3}\frac{\sin\theta}{r^2}\hat{\phi}$
coax	$A_z = -\frac{\mu_0 I}{2\pi}\ln\frac{s}{b}$ between; 0 out
sol.+axis wire	A_ϕ (sol.) + A_z (wire), superpose
$\mathbf{B} = kz\hat{x}$	$\mathbf{A} = -\frac{kz^2}{2}\hat{y}$ ($B_x = -\partial_z A_y$)
holed cyl, hole	$A_z = -\frac{\mu_0 J}{4}(2\mathbf{s}\cdot\mathbf{d} - d^2)$

M3 Cyl one-liners

$F(s)\hat{\phi}$: div 0, curl $\frac{1}{s}(sF)'\hat{z}$ · $F(s)\hat{z}$: div 0, curl $-F'\hat{\phi}$ · $F(s)\hat{s}$: div $\frac{1}{s}(sF)'$, curl 0 · $F(z)\hat{x}$: curl $F'\hat{y}$. Valid B: div=0. $\frac{\hat{\phi}}{s}$: curl = line current on axis.

Given B(s)φ̂ inside, out: $J_z = \frac{1}{\mu_0 s}(sB)'$; jump in B ⇒ K.

N · Dipoles & spinning charge

N1 Magnetic dipole

$\mathbf{m} = I\mathbf{a} = \frac{1}{2}\int\mathbf{r}\times\mathbf{J}\,d\tau$ · $\mathbf{B} = \frac{\mu_0 m}{4\pi r^3}(2\cos\theta\hat{r} + \sin\theta\hat{\theta}) = \frac{\mu_0}{4\pi r^3}[3(\mathbf{m}\cdot\hat{r})\hat{r} - \mathbf{m}]$

$\mathbf{N} = \mathbf{m}\times\mathbf{B}$ · $U = -\mathbf{m}\cdot\mathbf{B}$ · $\mathbf{F} = \nabla(\mathbf{m}\cdot\mathbf{B})$ · $U_{12} = \frac{\mu_0}{4\pi r^3}[\mathbf{m}_1\cdot\mathbf{m}_2 - 3(\mathbf{m}_1\cdot\hat{r})(\mathbf{m}_2\cdot\hat{r})]$ · coaxial $F = \frac{3\mu_0 m_1 m_2}{2\pi r^4}$

$m = \frac{Q}{2M}L$: ring $\frac{1}{2}Q\omega R^2$ · disk $\frac{1}{4}$ ($\frac{\pi}{4}\sigma\omega R^4$) · shell $\frac{1}{3}$ ($\frac{4\pi}{3}\sigma\omega R^4$) · ball $\frac{1}{5}$.

Center B (× $\frac{\mu_0 Q\omega}{\pi R}$): ring ¼ · disk ½ · shell ⅙ · ball ¼. Shell in: $\frac{2}{3}\mu_0\sigma\omega R$ uniform.

$\mathbf{B}_{avg}$ in sphere = $\frac{\mu_0}{4\pi}\frac{2\mathbf{m}}{R^3}$ (spin ball: $\frac{\mu_0 Q\omega}{10\pi R}$).

N2 Spinning solid ball (ρ, ω)

$\mathbf{B}_{in} = \mu_0\rho\omega[(\frac{R^2}{3} - \frac{r^2}{5})\cos\theta\hat{r} - (\frac{R^2}{3} - \frac{2r^2}{5})\sin\theta\hat{\theta}]$ = $\frac{\mu_0\rho\omega}{3}(R^2 - r^2)\hat{z} + \frac{\mu_0\rho\omega r^2}{15}(2\cos\theta\hat{r} + \sin\theta\hat{\theta})$. Center $\frac{\mu_0\rho\omega R^2}{3}$. Out: dipole $m = \frac{4\pi\rho\omega R^5}{15} = \frac{Q\omega R^2}{5}$.

Method: shells outside r give uniform $\frac{2}{3}\mu_0(\rho dr')\omega r'$; shells inside give dipole of $m(r)$.

N3 More moments & forces

Spinning cyl shell (σ, R, L): $m = \pi\sigma\omega R^3 L$.

Spinning solid cyl: $\frac{m}{L} = \frac{\pi\rho\omega R^4}{4}$. Spinning shell B on axis outside: $\frac{\mu_0 m}{2\pi z^3}$.

Dipole m (along $\hat{\phi}$) near wire I at s: $U = -\frac{\mu_0 Im}{2\pi s}$, $F_s = -\frac{\mu_0 Im}{2\pi s^2}$ (pulled in). m ∥ B in nonuniform B: pulled to strong B.

$J = J_0 e^{-s/a}$: $I_{enc} = 2\pi J_0 a^2[1 - (1 + \frac{s}{a})e^{-s/a}]$, $B = \frac{\mu_0 I_{enc}}{2\pi s}$. $J = \frac{k}{s}$: $I_{enc} = 2\pi ks$, B uniform $\mu_0 k$ inside.

O · Magnetic matter (M, H, µ)

O1 ★ Magnetization M, H

$\mathbf{J}_b = \nabla\times\mathbf{M}$ · $\mathbf{K}_b = \mathbf{M}\times\hat{n}$ · $\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$ · $\nabla\times\mathbf{H} = \mathbf{J}_f$ · $\oint\mathbf{H}\cdot d\mathbf{l} = I_f$ · $\nabla\cdot\mathbf{H} = -\nabla\cdot\mathbf{M}$

No free I + symmetry ⇒ H=0 ⇒ B=µ₀M. Free I: H by Ampère, $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$.

Linear: $\mathbf{M} = \chi_m\mathbf{H}$, $\mathbf{B} = \mu\mathbf{H}$, $\mu = \mu_0(1 + \chi_m)$, $\mathbf{J}_b = \chi_m\mathbf{J}_f$.

Uniform: sphere $\frac{2}{3}\mu_0\mathbf{M}$ in, $\mathbf{m} = \frac{4}{3}\pi R^3\mathbf{M}$, $\mathbf{H}_{in} = -\frac{\mathbf{M}}{3}$; long rod $\mu_0\mathbf{M}$; disk⊥ ≈0. $\mathbf{B}_{in} = \mu_0(1 - N)\mathbf{M}$, $\mathbf{H}_{in} = -N\mathbf{M}$. Shell: B=0 in cavity.

Linear sphere in $\mathbf{B}_0$: $\mathbf{B}_{in} = \frac{3\mu_r}{\mu_r+2}\mathbf{B}_0$, $\mathbf{M} = \frac{3}{\mu_0}\frac{\mu_r-1}{\mu_r+2}\mathbf{B}_0$. Superconductor $\chi_m = -1$: B=0 inside.

$M = ks^2\hat{\phi}$: $J_b = 3ks\hat{z}$, $K_b = -kR^2\hat{z}$, B=µ₀M in, 0 out. $M = ks\hat{z}$: $J_b = -k\hat{\phi}$, $K_b = kR\hat{\phi}$. $M = ks\hat{\phi}$: $J_b = 2k\hat{z}$. $M = kz\hat{x}$ slab: $J_b = k\hat{y}$, $K_b = -ka\hat{y}$ both faces.

Coax χm: $H = \frac{I}{2\pi s}$, $B = \frac{\mu_0(1+\chi_m)I}{2\pi s}$, $I_b = \chi_m I$ inner. Half-cyl $\mu_1|\mu_2$: $B = \frac{\mu_1\mu_2}{\mu_1+\mu_2}\frac{I}{\pi s}$. Toroid core: $B\times(1 + \chi_m)$.

Para: $\chi_m \sim 10^{-5} > 0$, M∥B, pulled in. Dia: $\chi_m < 0$, $\Delta m = -\frac{e^2R^2}{4m_e}B$, pushed out. Ferro: domains, hysteresis.

O2 H, M quick values

Uniform-M sphere: $\mathbf{H}_{in} = -\frac{\mathbf{M}}{3}$, $\mathbf{H}_{out} = \mathbf{B}_{out}/\mu_0$ (dipole). Long rod: $H_{in} \approx 0$, $B = \mu_0 M$. Thin disk: $H_{in} \approx -M$, $B \approx 0$. Bar-magnet pole density $\rho_M = -\nabla\cdot\mathbf{M}$, $\sigma_M = \mathbf{M}\cdot\hat{n}$ source H like charges.

B lines close (S→N inside); H lines start on N-face "poles", end on S.

O3 Linear magnetic shapes in B_0

$\mathbf{H}_{in} = \frac{\mathbf{H}_0}{1+N\chi_m}$, $\mathbf{B}_{in} = \mu\mathbf{H}_{in}$. Slab ⊥ (N=1): $B_{in} = B_0$, $H_{in} = H_0/\mu_r$. Slab ∥ (N=0): $H_{in} = H_0$, $B_{in} = \mu_r B_0$. Sphere: $\mathbf{B}_{in} = \frac{3\mu_r}{\mu_r+2}\mathbf{B}_0$. Cyl ⊥: $\mathbf{B}_{in} = \frac{2\mu_r}{\mu_r+1}\mathbf{B}_0$. Cavity in µ: $\frac{3}{2\mu_r+1}\mathbf{B}_0$ (B), sphere formula with $\mu_{in} \leftrightarrow \mu_{out}$.

P · BCs & E ↔ B analogies

P1 Magnetic BCs

$B^\perp$ cont. · $\mathbf{B}_{ab} - \mathbf{B}_{be} = \mu_0\mathbf{K}\times\hat{n}$ · $\mathbf{A}$ cont. · $\partial_n\mathbf{A}_{ab} - \partial_n\mathbf{A}_{be} = -\mu_0\mathbf{K}$ · $\mathbf{H}^\parallel$ jumps $\mathbf{K}_f\times\hat{n}$ · $\mu_1 H_1^\perp = \mu_2 H_2^\perp$ · $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_1}{\mu_2}$

P2 Electric ↔ magnetic

	P	M
bound	σ_b, ρ_b	K_b, J_b
uniform	caps	sides
sphere	$-\frac{\mathbf{P}}{3\epsilon_0}$	$+\frac{2}{3}\mu_0\mathbf{M}$
rod ∥	≈0	$\mu_0\mathbf{M}$
disk ⊥	$-\frac{\mathbf{P}}{\epsilon_0}$	≈0
aux	D ← ρ_f	H ← I_f
fill all	E ÷ εr	B × µr

P3 Electrostatics ↔ magnetostatics

ρ, ϵ_0	$\mathbf{J}$, $1/\mu_0$
$\nabla\cdot\mathbf{E} = \rho/\epsilon_0$	$\nabla\times\mathbf{B} = \mu_0\mathbf{J}$
$\nabla\times\mathbf{E} = 0$	$\nabla\cdot\mathbf{B} = 0$
$V = k\int\frac{\rho}{r}$	$\mathbf{A} = \frac{\mu_0}{4\pi}\int\frac{\mathbf{J}}{r}$
$\nabla^2 V = -\rho/\epsilon_0$	$\nabla^2\mathbf{A} = -\mu_0\mathbf{J}$
Gauss (symmetry)	Ampère (symmetry)
$\frac{p}{4\pi\epsilon_0}$	$\frac{\mu_0 m}{4\pi}$ (same field outside)
$E^\perp$ jumps σ/ϵ_0	$B^\parallel$ jumps $\mu_0 K$
V cont.	A cont.

P4 Useful vector facts

$\oint d\mathbf{l} = 0$ · $\oint\mathbf{r}\times d\mathbf{l} = 2\mathbf{a}$ · $\int_0^{2\pi}\hat{s}\,d\phi = 0$ · $\mathbf{m} = \frac{I}{2}\oint\mathbf{r}\times d\mathbf{l}$ · $\mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = \mathbf{b}(\mathbf{a}\cdot\mathbf{c}) - \mathbf{c}(\mathbf{a}\cdot\mathbf{b})$ · $\nabla(\mathbf{m}\cdot\mathbf{B}) = \mathbf{m}\times(\nabla\times\mathbf{B}) + (\mathbf{m}\cdot\nabla)\mathbf{B}$ (= $(\mathbf{m}\cdot\nabla)\mathbf{B}$ in current-free region).

Q · Exam I: separation & images

Q1 EXAM I · separation of variables

Cartesian $V = XY$: two zero walls in y ⇒ $\sin\frac{n\pi y}{a}$; x: $e^{-n\pi x/a}$ (to ∞), $\frac{\sinh(n\pi(b-x)/a)}{\sinh(n\pi b/a)}$ (zero at b), cosh (symmetric). $C_n = \frac{2}{a}\int_0^a V_0(y)\sin\frac{n\pi y}{a}dy$; const: $\frac{4V_0}{n\pi}$ odd. Slot: $V = \frac{4V_0}{\pi}\sum_{odd}\frac{1}{n}e^{-n\pi x/a}\sin\frac{n\pi y}{a}$. Two live faces: add. Box top+bottom V_0: $C_n\cosh\frac{n\pi}{2} = \frac{4V_0}{n\pi}$, center $\frac{V_0}{2}$; all faces $V_0 \Rightarrow V_0$. 3D: $\sinh(\pi\sqrt{(n/a)^2 + (m/b)^2}\,x)$, const: $C_{nm} = \frac{16V_0}{nm\pi^2}$ odd. Jump in edge value: series gives average.

Polar (long in z): $\Theta = \sin k\theta, \cos k\theta$; $R = r^{\pm k}$ (k=0: $\ln r$, θ). Walls θ=0,α grounded ⇒ $\sin\frac{n\pi\theta}{\alpha}$, $k = \frac{n\pi}{\alpha}$. Center ⇒ r^k; ∞ ⇒ r^{-k}; zero arc at a: $(\frac{r}{a})^k - (\frac{a}{r})^k$. One wall V_0: $V = \frac{V_0\theta}{\alpha}$. Wedge a<r<b, V₀ on b: $C_n = \frac{4V_0}{n\pi[(b/a)^k - (a/b)^k]}$. Full circle: k = 1,2,…

Legendre: $\cos^2\theta = \frac{1}{3} + \frac{2}{3}P_2$; $\cos^3\theta = \frac{3}{5}P_1 + \frac{2}{5}P_3$; $c_l = \frac{2l+1}{2}\int_{-1}^1 fP_l dx$. Shell at $V(\theta) = \sum c_l P_l$: in $c_l(\frac{r}{R})^l P_l$, out $c_l(\frac{R}{r})^{l+1}P_l$, $\sigma = \frac{\epsilon_0}{R}\sum(2l+1)c_l P_l$. Shell $\sigma = \sum\sigma_l P_l$: $A_l = \frac{\sigma_l}{(2l+1)\epsilon_0 R^{l-1}}$, $B_l = A_l R^{2l+1}$. $V_0\cos^2\theta$ shell: in $V_0[\frac{1}{3} + \frac{2}{3}\frac{r^2}{R^2}P_2]$, out $V_0[\frac{R}{3r} + \frac{2}{3}\frac{R^3}{r^3}P_2]$, $\sigma = \frac{\epsilon_0 V_0}{R}[\frac{1}{3} + \frac{10}{3}P_2]$. $\sigma = k\cos\theta$: $A = \frac{k}{3\epsilon_0}$, uniform E inside $\frac{k}{3\epsilon_0}$ down.

Q2 EXAM I · images & superposition

Plane: σ total $-q$, half within $s = \sqrt{3}d$. q inside grounded shell at d: image $-\frac{qR}{d}$ at $\frac{R^2}{d}$; $F = \frac{kq^2Rd}{(R^2-d^2)^2}$ toward wall. Sphere at V_0 + q at a: images $-\frac{qR}{a}$ at $\frac{R^2}{a}$ & $4\pi\epsilon_0 RV_0$ at center; $F = \frac{qRV_0}{a^2} - \frac{kq^2Ra}{(a^2-R^2)^2}$. Neutral: sphere's $V = \frac{kq}{a}$, always attracts. Corner 90° at (x_0, y_0): $-q(-x_0, y_0)$, $-q(x_0, -y_0)$, $+q(-x_0, -y_0)$; 60°: 5 images. Images valid only outside metal.

Cylinder (+ρ, radius a, along z) through ball (−ρ, R): inside both: $\frac{\rho s}{6\epsilon_0}$ out, $\frac{\rho z}{3\epsilon_0}$ in; plane z=0: $\frac{\rho a^2}{2\epsilon_0 s} - \frac{\rho s}{3\epsilon_0}$. Hole = whole + negative copy, each from its own center.

R · Answers & strategy

R1 HW answers: magnetism

5.4 $\mathbf{F} = Ika^2\hat{z}$. **5.6** $K = \sigma\omega r$; $\mathbf{J} = \frac{3Q\omega}{4\pi R^3}r\sin\theta\hat{\phi}$. **5.8** $\frac{\sqrt{2}\mu_0 I}{\pi R}$; $\frac{n\mu_0 I}{2\pi R}\sin\frac{\pi}{n} \to \frac{\mu_0 I}{2R}$. **5.10a** $\frac{\mu_0 I^2 a^2}{2\pi s(s+a)}$ toward. **5.11** $\frac{\mu_0 nI}{2}(\cos\theta_2 - \cos\theta_1)$; ∞: $\mu_0 nI$. **5.14** (a) 0 in (b) $\frac{\mu_0 Is^2}{2\pi a^3}$; out $\frac{\mu_0 I}{2\pi s}$. **5.15** $-\mu_0 Jz\hat{y}$, $\mp\mu_0 Ja\hat{y}$. **5.17** $\mu_0 I(n_1 - n_2)$, $\mu_0 In_2$, 0. **5.18** $\mu_0\sigma v$; $\frac{\mu_0\sigma^2v^2}{2}$ up (repel); $v = c$. **5.25** $\frac{\mu_0 I}{4\pi}\ln\frac{z_2+\sqrt{z_2^2+s^2}}{z_1+\sqrt{z_1^2+s^2}}$; curl → $\frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$. **5.26** $\mathbf{J} = \frac{k}{\mu_0 s^2}\hat{\phi}$. **5.29** $-\frac{\mu_0 K}{2}|z|$ along K. **5.37** $m = I\pi R^2$; $B \to \frac{\mu_0 IR^2}{2z^3}$. **5.39** disk $\frac{\pi}{4}\sigma\omega R^4$; shell $\frac{4\pi}{3}\sigma\omega R^4$. **5.63** $m = \frac{Q\omega R^2}{5}$; $B_{avg} = \frac{\mu_0 Q\omega}{10\pi R}$; $\mathbf{A} \approx \frac{\mu_0 Q\omega R^2\sin\theta}{20\pi r^2}\hat{\phi}$ (exact outside). **6.3** $\frac{3\mu_0 m_1 m_2}{2\pi r^4}$. **6.8** $J_b = 3ks\hat{z}$, $K_b = -kR^2\hat{z}$, $B = \mu_0 ks^2\hat{\phi}$ in, 0 out. **6.12** $J_b = -k\hat{\phi}$, $K_b = kR\hat{\phi}$, $B = \mu_0 ks\hat{z}$ in. **6.16** $B = \frac{\mu_0(1+\chi_m)I}{2\pi s}$, $M = \frac{\chi_m I}{2\pi s}$, $K_b = \frac{\chi_m I}{2\pi a}\hat{z}$ (a), $-\frac{\chi_m I}{2\pi b}\hat{z}$ (b), $J_b = 0$.

R2 Mock answers to remember

Ball of εr, $\rho_f = kr$: $V(0) = \frac{kR^3}{4\epsilon_0}(1 + \frac{1}{3\epsilon_r})$, $W = \frac{\pi k^2R^7}{8\epsilon_0}(1 + \frac{1}{7\epsilon_r})$. $\epsilon = \epsilon_0\frac{b}{r}$ shell: $C = \frac{4\pi\epsilon_0 b}{\ln(b/a)}$. $\epsilon = \epsilon_0(1 + \frac{r}{a})$: $V = kQ[\frac{1}{a} - \frac{1}{b} + \frac{1}{a}\ln\frac{a+b}{2b}]$. Hole b=a/2, d=a/3: $I_\pm = \frac{4}{3}I, -\frac{1}{3}I$, $B_{hole} = \frac{2\mu_0 I}{9\pi a}$. P=ks ŝ + axis λ: $E = \frac{1}{\epsilon_0}(\frac{\lambda}{2\pi s} - ks)$. $\mathbf{E} = kr^3$ ball: $W = \frac{20\pi\epsilon_0 k^2R^9}{9}$.

R3 50 minutes

Read all → write standard results → easy parts first → setup + BCs = most points → 3-min stall rule → check limits & units (T = $\mu_0 I/L = \mu_0 K = \mu_0 M$).